Insiders and Outsiders in the Rental Market: Entry Rents, Incomes and Housing Supply in Spain

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Abstract

Spanish tenants who stay in their homes have seen rents grow more slowly than local incomes: between 2015 and 2023, the official rent index for all leases in force rose by 16.5 log points and median income per consumption unit by 31.0, and the rent-to-income ratio fell in every one of the 703 municipalities covered. Households signing a new lease faced different prices. In 2024, new leases cost 15.9 log points more than leases in force per euro of cadastral value, in 99.5 % of the larger municipalities; in Catalonia and Aragón, rents at signature outpaced incomes by 11.7 and 9.5 log points since 2015. A stock-flow model with long leases, indexed updates and rent resets explains this insider–outsider divergence and predicts that local demand shocks load on entry rents, that incumbents stay while would-be entrants are priced out, and that supply constraints amplify the reset. Using a shift-share instrument for immigration-driven population growth, I find that a cumulative inflow of 1 % of the population raises the quality-adjusted entry gap by 0.57 % and implied entry rents by 0.83 %, against 0.22 % for all leases; income per consumption unit does not respond, household size rises and fewer Spanish nationals move into the municipality, while none move out. The entry gap responds more where land is constrained by sea or slope. Tourism moves dwellings across the same entry margin above a threshold of about 2.4 % of the stock, and rent caps on new leases shift contracts towards unregulated seasonal lets. Spain's rental affordability problem is a problem of access.

Keywords Rental housing · Affordability · Entry rents · Housing supply · Immigration · Spain

JEL Classification R21 · R31 · R38 · J61

1 Introduction

Rents top the list of economic concerns in Spain, yet the official index of rents paid by tenants has grown no faster than their incomes. Between 2015 and 2023 the INE rent index, which covers all leases declared in income-tax returns, rose by 16.5 log points; median household income per consumption unit rose by 31.0. In none of the 703 municipalities with an official index did rents outpace incomes. The rent faced by a household signing a lease tells a different story. In 2024, new leases cost 15.9 log points more than leases in force for dwellings of the same cadastral value, and the gap is positive in 99.5 % of the 402 municipalities with more than 20,000 inhabitants. In Catalonia, where every new lease is registered, rents on new contracts rose by 39.0 log points between 2015 and 2023, 11.7 more than incomes; in Zaragoza, rents at signature rose by 37.6 log points for dwellings on the same street, against 14.9 for all leases and 26.1 for incomes.

This paper argues that the Spanish rental affordability problem is a problem of entry, explains why it arises and identifies the forces that make it larger in some markets than in others. The starting point is that the average rent of all leases, the object of official statistics, is a poor measure of the cost of finding housing today. Leases last at least five years, existing rents are updated by the consumer price index or by a legal cap, and only a small and falling share of leases is renewed each year: about one in six. In such a market the adjustment to demand shocks falls on the rent reset at the start of a new contract. Incumbent tenants are insulated; households that move, form or arrive pay the market price. The affordability of insiders and the affordability of outsiders diverge.

Section 4 formalises this idea in a stock-flow model of a local rental market with contract duration, indexed updates and rent resets, in which the mobility, incomes and size of households respond to the same shocks. The model delivers five predictions. A demand shock raises entry rents more than the average rent of all leases, which absorbs it only at the rate at which leases turn over. Whether the shock worsens affordability depends on whether it raises the income of those who enter, which a measure of income per household can conceal when households grow larger. The response of entry rents is larger where new supply and the reallocation of dwellings from other uses are inelastic. Because incumbents gain from staying as the gap widens, turnover falls and quantities adjust through the households who do not get in rather than through those who leave. Finally, rules that protect incumbents widen the gap between entry rents and the rents of sitting tenants, while caps on entry rents narrow it at the cost of fewer new leases.

The empirical analysis tests these predictions with open administrative data. Tax statistics give, for 2024, the rent, floor area and cadastral value of new leases and of all leases in every municipality above 20,000 inhabitants and in the postal codes of the largest cities, which allows a quality-adjusted measure of the entry gap. Deposit registers in Catalonia, Aragón, the Basque Country and the Comunitat Valenciana

record rents at signature; the Aragón register, with 458,392 residential deposits since 1996 and the street of each dwelling, also reveals how long leases last. The household income atlas gives income per consumption unit and household size by municipality, the household budget survey gives the rent burden of renters, and the statistics of residential changes give moves between municipalities. The demand shock is the immigration-driven inflow of residents predicted by a shift-share instrument based on the settlement of 33 origin groups in 2003 and their national growth (Saiz 2007; Goldsmith-Pinkham et al. 2020; Borusyak et al. 2022), taken from a companion paper (Fernández-Aguilera 2026).

Six results stand out. First, a cumulative inflow of 1 % of the population raises the quality-adjusted entry gap by 0.57 % (Anderson-Rubin set [0.11, 1.74]) and implied entry rents by 0.83 % [0.25, 2.21], while the effect on the rent of all leases is 0.22 and not significant. Second, the same shock does not raise income per consumption unit (0.08), but it raises household size (1.26) and therefore income per household (0.89): measured per household, affordability would appear to improve because more adults share each dwelling. Third, the shock does not push incumbents out: the out-migration of Spanish nationals to other municipalities does not change (-0.01), while their in-migration falls by -0.54 moves per 100 residents per point of inflow [-2.00, -0.14]. Adjustment in quantities, like adjustment in prices, falls on entrants. Fourth, the dwelling stock does not respond, and the entry gap responds more to demand where land is constrained by sea or steep slopes (0.49 per standard deviation, $p = 0.019$) and in coastal municipalities. Fifth, tourism moves dwellings across the same entry margin above a threshold of about 2.4 % of the stock in tourist use: when tourism collapsed in 2020–21, more long-term leases were signed above that threshold, and rents on new leases rose with the recovery. Sixth, regulation acts at the margin the model predicts: the caps on updates of 2022–2024 account for up to 58 % of the widening of the national entry gap, and the Catalan cap on new-contract rents lowered the number of regulated leases and raised the share of unregulated seasonal leases by 1.8 percentage points.

The paper contributes to three literatures. It adds to the economics of rental affordability (Quigley and Raphael 2004; Khametshin et al. 2024) a distinction between insiders and outsiders, a model of why they diverge in markets with long indexed leases, and evidence that they do. It contributes to the measurement of rents by showing why indices of all leases (Genesove 2003; Ambrose et al. 2015) understate what entrants pay, by how much and for how long, and by constructing a quality-adjusted entry gap from tax data. And it links the literatures on local demand shocks and supply constraints (Saiz 2010; Glaeser and Gyourko 2018; Hilber and Vermeulen 2016) and on rent regulation (Autor et al. 2014; Diamond et al. 2019; Jofre-Monseny et al. 2023) through the margin at which adjustment takes place: in markets that protect incumbents, both prices and quantities adjust at entry.

2 Institutions and measurement

Leases and updates. Since March 2019 residential leases in Spain last at least five years when the landlord is a natural person and seven when it is a company, and annual updates of existing leases are usually tied to the consumer price index. Updates were capped at 2 % from the end of March 2022 to the end of 2023 and at 3 % in 2024, and since 2025 they follow a reference index published by INE. Rents are set freely at the start of a new lease except in markets declared tensioned under the 2023 Housing Act, where they are capped by the rent of the previous lease and a reference index; Catalonia applied this cap in 140 municipalities from 16 March 2024 and in 131 more from October 2024. Seasonal leases, for stays of less than a year for work, study or other temporary reasons, are exempt.

Measuring rents. The INE index (IPVA) averages the rents of all leases in force declared in income-tax returns, excluding the Basque Country and Navarre. Since 2021 INE also publishes separate indices for new and existing leases; the implied weight of new leases was 17.6 % in 2021 and 12.3 % in 2024. The tax agency publishes, for 2024, the average rent, floor area and cadastral reference value of all leases and of leases signed in the year, by municipality above 20,000 inhabitants and by postal code in the largest cities. Deposit registers record leases at signature in Catalonia (mean rent, municipality and quarter, since 2005), Aragón (one record per deposit with rent, street and whether the deposit has been returned, since 1996), the Basque Country (rent per m^2 built, municipality and year, since 2016) and the Comunitat Valenciana (deposit amount, since 2020).

3 Related literature

Rental affordability is usually measured as the ratio of rent to the income of renters (Quigley and Raphael 2004). For Spain, Khametshin et al. (2024) document the rise in rental demand since 2015 among young and foreign-born households and in large urban and tourist areas, and the high rent burden of low-income tenants, and describe the quantification of its causes as an open question. Indices of rents on all leases move slowly because existing rents are sticky (Genesove 2003) and because tenants who stay pay less than new ones (Ambrose et al. 2015). Rent control turns that wedge into a policy instrument: it benefits incumbent tenants, lowers their mobility and reduces the supply of rental housing to newcomers (Autor et al. 2014; Diamond et al. 2019; Kholodilin 2024), and in Catalonia it lowered rents and the number of leases signed (Jofre-Monseny et al. 2023; Pérez García 2026).

Local demand shocks are capitalised into rents where supply is inelastic (Roback 1982; Saiz 2010; Glaeser and Gyourko 2018; Hilber and Vermeulen 2016; Hsieh and Moretti 2019). Immigration is the standard source of such shocks: Saiz (2007) finds that an inflow of 1 % of a city's population raises rents by about 1 %, and Gonzalez and Ortega (2013) and Sanchis-Guarner (2023) study house prices in Spain. Shift-share designs identify such effects either from the exogeneity of the shocks (Borusyak et al. 2022;

Adão et al. 2019; Borusyak and Hull 2023) or from that of the initial shares (Goldsmith-Pinkham et al. 2020), and immigration instruments based on past settlement can confound current and past inflows (Jaeger et al. 2018). Short-term rentals reallocate dwellings from residents to visitors (Barron et al. 2021); in Barcelona they raised rents (Garcia-López et al. 2020), and in Lisbon the collapse of tourism in 2020 lowered rents and increased the long-term rental supply (Batalha et al. 2022). None of these strands separates the rents paid by insiders from those faced by entrants, which is the focus of this paper.

4 A model of insiders, entrants and rent resets

The model describes a local rental market i in which leases are long, existing rents follow an update rule and a market price applies only when a dwelling is relet. It has three blocks: an accounting block that links the rents of all leases to entry rents through turnover; a market block that determines entry rents, incomes per consumption unit and household size; and a behavioural block in which turnover itself depends on the gap between the two rents. Throughout, lower-case letters are logs.

4.1 Leases, resets and the index of all leases

In year t a share λ_{it} of leases ends and the dwelling is relet at the market-clearing entry rent R^*_{it}; the remaining leases are updated by $u_t = \min(\pi_t, \bar{u}_t)$, inflation or a legal cap. The log of the average rent of all leases, $\bar{r}_{it}$, then satisfies

$$\bar{r}_{it} = \lambda_{it}\, r^*_{it} + (1-\lambda_{it})\,(\bar{r}_{i,t-1} + u_t), \tag{1}$$

and the entry gap $G_{it} \equiv r^*_{it} - \bar{r}_{it}$ evolves as

$$G_{it} = (1-\lambda_{it})\left[G_{i,t-1} + \Delta r^*_{it} - u_t\right]. \tag{2}$$

The gap widens whenever entry rents grow faster than the update rule, and it is more persistent the lower the turnover λ. Two implications follow. First, if entry rents grow at a constant rate g and updates at u, the gap converges to

$$G^* = \frac{1-\lambda}{\lambda}\,(g-u). \tag{3}$$

With the turnover measured in section 6, $(1-\lambda)/\lambda$ is between 4 and 7: each point by which entry rents outgrow the update rule sustains a gap of 4 to 7 points. Second, a permanent shock to entry rents passes to the index of all leases at rate λ: the index absorbs a share $1-(1-\lambda)^k$ of the shock after k years, about one sixth after one year and a half after four. An index of all leases therefore lags the market, and the lag is longer where leases are longer. Longer minimum durations lower λ; caps on updates lower u_t. Both widen the gap for a given path of entry rents. A cap on entry rents limits Δr^* directly.

4.2 Entry rents, incomes and household size

Entry rents clear the flow market for new leases. Demand comes from households that form, move or arrive; their number grows with the local population N_{it}, and each consumes $q_{it} = A\,(Y_{it}/S_{it})^{\alpha} R_{it}^{*\,-\eta}$ units of housing per person, where Y_{it}/S_{it} is income per consumption unit, S_{it} an equivalence scale that increases with household size, and η the price elasticity of space per person. A high η means that higher rents are absorbed by sharing dwellings. Supply comes from dwellings released by sitting tenants, new construction with elasticity σ_i, and dwellings reallocated from tourist use or non-use with elasticity ε_i. Log-linearising, a population shock $d\ln N_{it}$ raises entry rents by

$$d\,r_{it}^{*} = \frac{d\ln N_{it} + \alpha\,(d\ln Y_{it} - d\ln S_{it})}{\eta + \varepsilon_i + \sigma_i}, \tag{4}$$

and the affordability of entrants, $A_{it}^{E} = r_{it}^{*} - \ln(Y_{it}/S_{it})$, changes by

$$dA_{it}^{E} = d\,r_{it}^{*} - \big(d\ln Y_{it} - d\ln S_{it}\big). \tag{5}$$

The affordability of insiders, $A_{it}^{I} = \bar{r}_{it} - \ln(Y_{it}/S_{it})$, changes only by $\lambda\,d\,r_{it}^{*}$ on impact. When the shock comes from arrivals with incomes below the local mean, $d\ln Y_{it} - d\ln S_{it}$ is small or negative, η shows up as larger households, and income per household, $d\ln Y_{it}$, rises with household size even if income per consumption unit does not. Tourism enters through ε_i. Tourist demand withdraws dwellings from long-term letting and a collapse of tourist demand returns them; but the stock that can be returned is small where tourist lets are rare, so that ε_i is close to zero below some share of dwellings in tourist use and positive above it.

4.3 Turnover, lock-in and who adjusts

Turnover is not a constant. A sitting tenant who moves within the market gives up a rent that is G below the market, so the probability of moving falls with the gap: $\lambda_{it} = \lambda_0 - \kappa G_{it}$ with $\kappa > 0$, as in the lock-in effects of rent control (Diamond et al. 2019). Substituting into (2), a shock that widens the gap also lowers turnover, which makes the gap more persistent and reduces the number of dwellings that come back to the market each year. The flow of vacancies available to entrants shrinks exactly when entry demand rises. Because incumbents are insulated from the reset, a demand shock does not push them out; it rations those who would have entered. In a market with many local municipalities, the households priced out at entry move elsewhere or do not move at all, and the local in-migration of other entrants falls.

4.4 Regulation

Equation (2) gives the mechanical effect of a cap on updates: it lowers u_t and widens the gap one for one for the leases updated while it binds. A forward-looking landlord adds a second effect. If the landlord sets the entry rent so that the expected real rent over the life of the lease is unchanged, and expects updates to fall short of inflation by $\pi - \bar{u}$ in each year of the lease, the entry rent rises by approximately $(\pi - \bar{u})\,\bar{k}$, where $\bar{k} = \beta(1-\lambda)/[1-\beta(1-\lambda)]$ is the discounted expected time elapsed within a lease and β the discount factor. With $\lambda = 0.18$ and $\beta = 0.96$, $\bar{k}$ is about 3.7 years. A permanent cap therefore raises entry rents by front-loading; a temporary cap, like the Spanish one, mainly widens the gap mechanically. A cap on entry rents lowers r^* in the regulated segment but lowers the return on letting there, so owners who can move dwellings to unregulated uses, such as seasonal lets, do so, and the number of regulated leases falls.

4.5 Predictions

P1 (entry loading). A demand shock raises entry rents by more than the rent of all leases, which absorbs it at the rate λ, so that the entry gap G widens.

P2 (income measurement). A shock that raises household size S raises income per household mechanically without raising income per consumption unit. Affordability measured per household can improve while the affordability of entrants deteriorates.

P3 (amplification). The response of entry rents is larger where new supply (σ_i) and reallocation (ε_i) are less elastic: in land-constrained markets and where few dwellings can be moved from other uses into long-term letting. Reallocation from tourist use is non-linear in the share of tourist lets.

P4 (who adjusts). Incumbents do not leave in response to a demand shock; quantities adjust through lower in-migration of other would-be entrants, and turnover falls as the gap widens.

P5 (regulation). Caps on updates widen the gap; caps on entry rents narrow it and reduce the number of regulated leases, shifting lets towards unregulated uses.

4.6 Incidence

The model also says who bears the cost of a demand shock. With λ the share of households that sign a lease in a year, the extra rent paid in that year is $\lambda\, d\, r^*$ per dwelling, all of it by entrants, while sitting tenants pay only the update. As long as the gap persists, the burden of the shock is shared between entrants and landlords, and sitting tenants gain relative to the market. Affordability statistics based on all leases or on all renters average the two groups with weights λ and $1-\lambda$ and therefore show only about a sixth of the change faced by entrants in the year of the shock.

5 Data

Entry gap. The main outcome is the quality-adjusted entry gap in 2024 from the tax statistics of dwellings declared in income-tax returns:

$$G_i^{VR} = \ln\Big(\frac{\text{rent}_i^{new}}{\text{value}_i^{new}}\Big) - \ln\Big(\frac{\text{rent}_i^{all}}{\text{value}_i^{all}}\Big),$$

where value is the mean cadastral reference value of the dwellings. G^{VR} equals the log ratio of gross yields on new and on all leases. A version per m^2 and the raw gap are used for robustness. In Spain as a whole, new leases averaged 816 euros per month against 729 for all leases, while new leases were on dwellings of lower value (145,067 against 154,022 euros): the gross yield was 6.8 % against 5.7 %. The same statistics by postal code cover 318 postal codes in 15 cities.

Rents, incomes and moves over time. INE indices for all, new and existing leases; deposit registers in Catalonia, Aragón, the Basque Country and the Comunitat Valenciana; the INE household income atlas (2015–2023) for median net income per consumption unit, income per person and per household, and average household size; the INE household budget survey (2016–2024) for the rents, incomes and floor area of renting households (23,274 household-years); and the INE statistics of migrations and changes of residence for moves between municipalities by nationality (2021–2024). In Aragón, I match 99.95 % of residential deposits to municipalities and build a composition-adjusted index of entry rents from a regression of log rent on year effects and street fixed effects within each municipality.

Demand shock and controls. The cumulative net inflow of foreign-born residents between 2015 and 2024 relative to 2015 population, from the municipal register (to 2021) and the annual census (from 2021), using within-source changes. The instrument sums the companion paper's annual shift-share predictions over the same years. Controls are predetermined characteristics from the 2011 census and 2012 firm counts.

Supply constraints, tourism, regulation and tenure.

- Supply constraints: share of land within 10 km of each municipality that is sea or has a slope above 15 %, computed from the Copernicus 90-metre digital elevation model in the spirit of Saiz (2010). Cadastral dwelling counts measure the supply response.
- Tourism: tourist dwellings as a share of all dwellings (INE, August 2020).
- Regulation: seasonal leases by municipality and quarter in Catalonia (2023–2026); CPI and the legal caps on updates.
- Tenure: appraised housing values by municipality (Ministry of Housing) relative to income, as a measure of exposure to the rise in mortgage rates.

All data are public and are downloaded by the replication code.

6 Insiders and entrants

Table 1 and figure 1 set out the central facts. For all leases, rents grew by 16.5 log points between 2015 and 2023 and income per consumption unit by 31.0: the rent-to-income ratio of sitting tenants fell by −14.4 points on average, by −19.5 in municipalities of 10,000–20,000 inhabitants and by −7.9 in the largest cities. Not a single municipality shows the opposite. The household budget survey confirms this for renters as a whole: their median rent-to-income ratio was 27.6 % in 2016 and 26.4 % in 2024, and the share paying more than 40 % of income fell from 25 % to 19 %.

New leases tell a different story in three of the four regions with registers. In Catalonia, rents on new leases rose by 39.0 log points against 27.3 for incomes, and they outpaced incomes in 77 % of municipalities. The gap between entry rents and incomes increases with the size of the municipality (figure 1b). In Aragón, where the street fixed effects hold the location of the dwelling constant, rents at signature rose by 36.1 log points against 26.5 for incomes and 15.1 for all leases; they outpaced incomes in 77 % of the 43 municipalities with enough leases and outpaced the index of all leases in every one of the 14 that have one (figure 3a). In the Comunitat Valenciana, deposits on new leases rose by 22.9 points between 2020 and 2023, against 15.9 for incomes. In the Basque Country, however, rents per m^2 on new leases rose by 21.2 points between 2016 and 2023, less than incomes (26.4), and they outpaced incomes in only 10 % of municipalities: the divergence at entry is not universal.

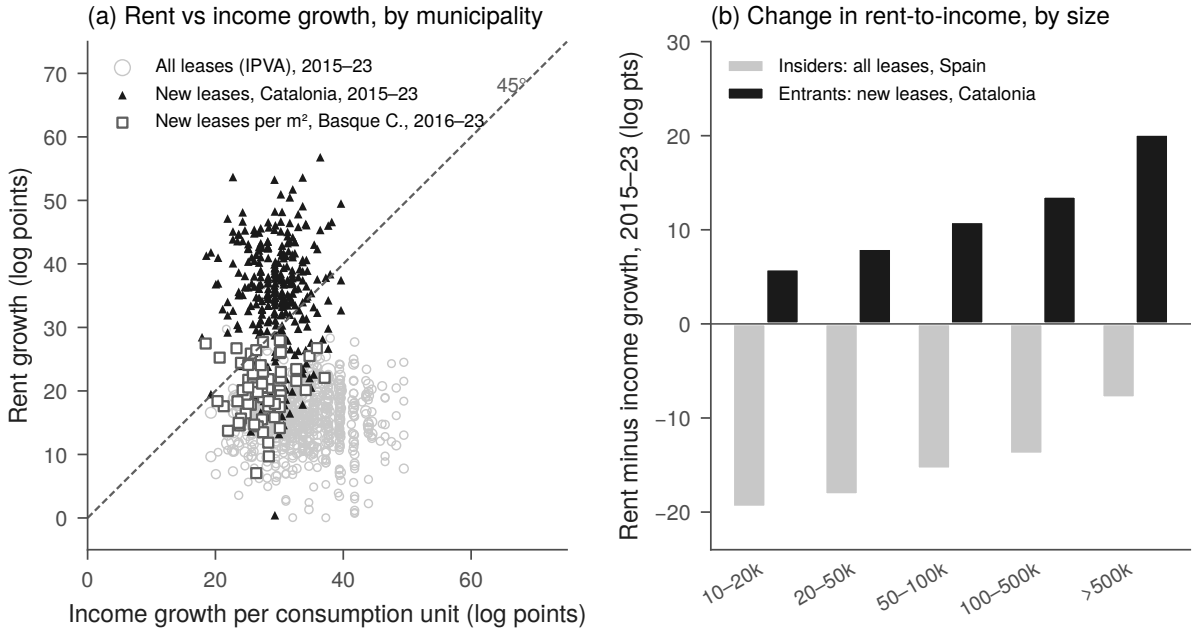


Fig. 1 Rents of insiders and entrants against incomes

Notes: panel (a), growth of rents and of median income per consumption unit by municipality: index of all leases (IPVA, 2015–2023), mean rent of new leases in Catalonia (2015–2023, municipalities with at least 30 leases in 2015) and rent per m^2 of new leases in the Basque Country (2016–2023). Panel (b), population-weighted difference between rent and income growth by size of municipality. Sources: INE, Generalitat de Catalunya, Gobierno Vasco.

The entry gap is widening nationally. Relative to 2015, the INE index of new leases exceeded the

Table 1 Rents of insiders and entrants against incomes

Rent measure	Period	Units	Rent growth	Income growth	Difference	Share with rent > income
			(log points, weighted)			
All leases (IPVA), Spain	2015–2023	703 municipalities	1650.0	31.0	1619.0	0 %
New leases (deposits), Catalonia	2015–2023	289 municipalities	39.0	27.3	11.7	77 %
New leases per m^2 (deposits), Basque Country	2016–2023	67 municipalities	21.2	26.4	−5.2	10 %
New leases, street-adjusted index (deposits), Aragón	2015–2023	43 municipalities	36.1	26.5	9.5	77 %
Deposits on new leases, Comunitat Valenciana	2020–2023	104 municipalities	22.9	15.9	7.0	–
Entry gap: new leases relative to leases in force						
INE index, new vs existing, Spain	2021–2024	national	17.6	7.2	10.4	
AEAT, new vs all leases, per euro of cadastral value	2024	402 municipalities			15.9	99.5 %[a]

Notes: income is the median net income per consumption unit of the municipality (INE household income atlas). Means weighted by 2015 population. IPVA: index of all leases in force (tax returns). Deposit registers record leases at signature: mean rent in Catalonia (contracts of more than one year), rent per m^2 built in the Basque Country (free-market collective main residences), an index of rents per contract with street fixed effects in Aragón, deposit amount in the Comunitat Valenciana (one month of rent by law; partial coverage). AEAT: tax statistics of dwellings declared in 2024, municipalities above 20,000 inhabitants; the entry gap is $\ln(\text{rent}_{new}/\text{value}_{new}) - \ln(\text{rent}_{all}/\text{value}_{all})$. [a] Share of municipalities with a positive gap.

index of existing leases by 2.3 log points in 2021 and by 12.7 in 2024, and the gap widened in every province (figure 2a). In levels, the tax data show a quality-adjusted gap of 15.9 log points in 2024, with a median of about 15 points in every size class (figure 2b). New leases are signed on dwellings of lower value and similar size; the raw gap (13.9) understates the quality-adjusted one. Within the largest cities, the gap is positive in 99.4 % of postal codes (mean 17.1 points); 65 % of its variance is within cities, but it is unrelated to the rent level of the neighbourhood (slope −0.011, standard error 0.015). The entry gap is a feature of the market, not of particular neighbourhoods.

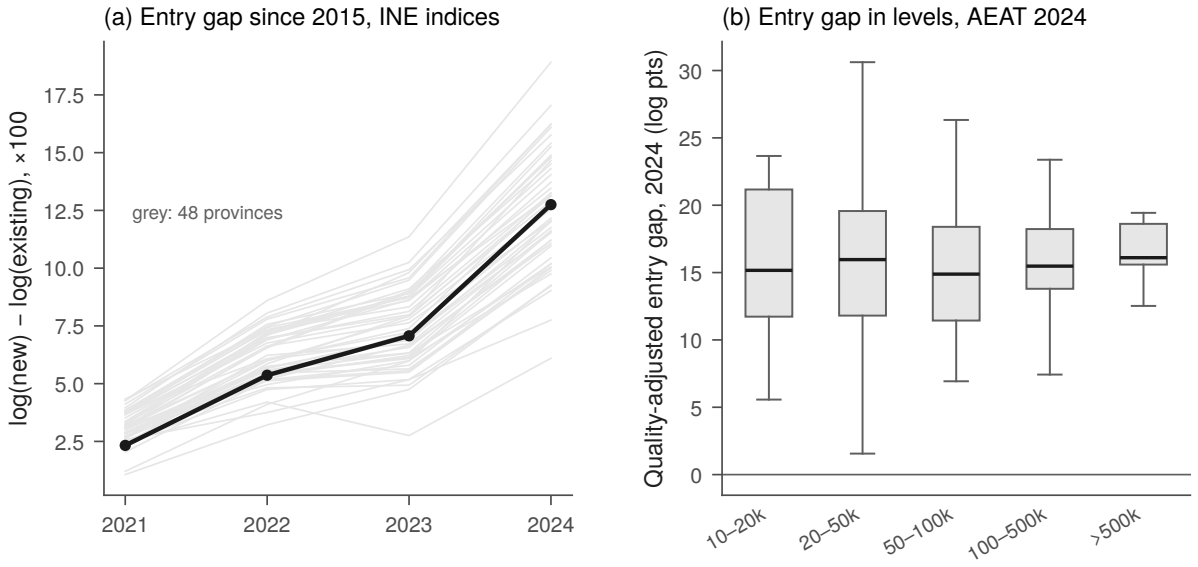


Fig. 2 The entry gap over time and in levels

Notes: panel (a), $100 \times \ln(\text{index}_{new}/\text{index}_{existing})$, both indices with base 2015 = 100 (INE table 59005), Spain and 48 provinces. Panel (b), quality-adjusted entry gap in 2024 (AEAT) by size of municipality: median, interquartile range and 1.5 interquartile-range whiskers.

How fast leases turn over determines how slowly the index of all leases follows the market. The Aragón register shows, for each cohort of deposits, the share not yet returned. A survival curve with

a constant hazard fits these shares closely (figure 3b) and implies that 19.2 % of leases end each year (15.7 % in Zaragoza), with a median duration of 3.3 years. This is close to the weight of new leases implied by the INE indices, 17.6 % in 2021, which fell to 12.3 % in 2024 as the gap widened, as the lock-in mechanism of section 4 predicts. With λ between 0.12 and 0.19, equation (3) implies that entry rents growing two points a year faster than updates sustain a gap of 8 to 14 points, the order of magnitude observed in 2024.

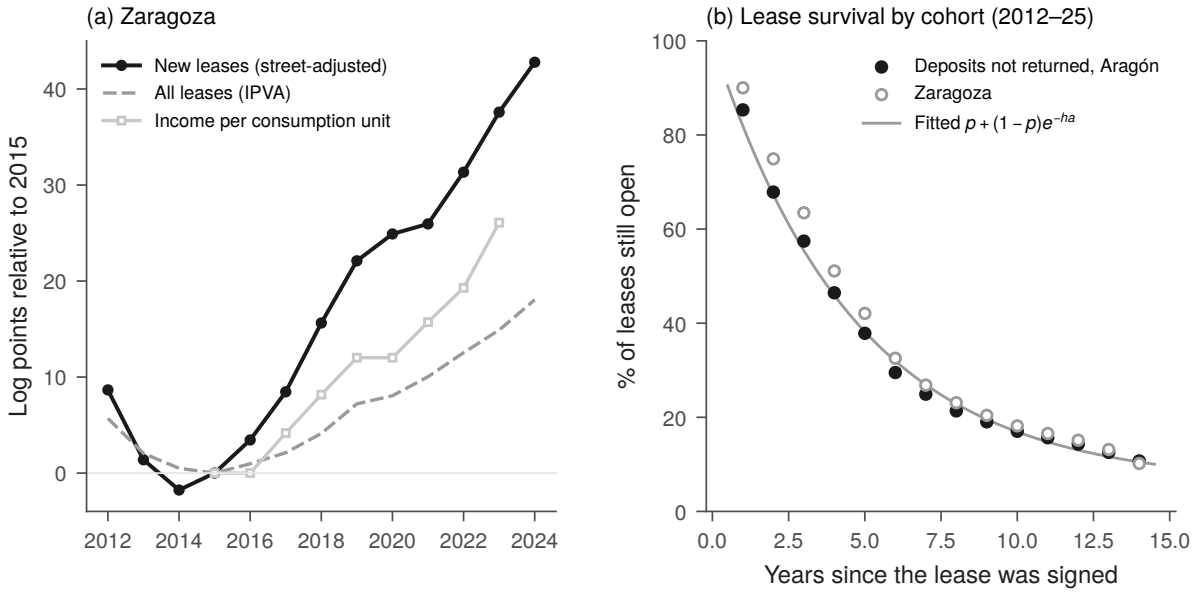


Fig. 3 Entry rents and lease turnover in the Aragón deposit register

Notes: panel (a), Zaragoza: index of rents of new leases with street fixed effects (Aragón deposit register), IPVA index of all leases and median income per consumption unit, log points relative to 2015. Panel (b), share of each cohort of residential deposits (2012–2025) not returned at the time of download, Aragón and Zaragoza, and fitted survival curve $S(a) = p + (1-p)e^{-ha}$ for Aragón. Sources: Gobierno de Aragón, INE.

7 Empirical strategy

For municipality i in province p,

$$y_i = \beta\, x_i + \boldsymbol{w}_i'\boldsymbol{\gamma} + \delta_p + \varepsilon_i, \tag{6}$$

where x_i is the cumulative foreign-born inflow relative to 2015 population and y_i is, in turn, the entry gap, the growth of the rent of all leases, implied entry-rent growth (the sum of the two), income per consumption unit, income per person and per household, household size, population, the dwelling stock and moves between municipalities. Affordability outcomes are rent growth minus growth of income per consumption unit, for insiders (all leases) and for entrants (implied entry rents). x_i is instrumented with the cumulative shift-share prediction $z_i = \sum_t \sum_o \ell_{io}\,(g_{ot} - g_{opt})/P_i$, where ℓ_{io} is municipality i's share of the 2003 national stock of origin o and $g_{ot} - g_{opt}$ the national change in that stock excluding province p. Regressions are weighted by 2015 population, standard errors are clustered by province and Anderson-Rubin sets are reported throughout. The pre-period falsification regresses rent growth in 2012–2015 on

the 2015–2024 inflow.

The instrument can be justified by the exogeneity of the origin shocks (Borusyak et al. 2022) or by that of the 2003 settlement shares (Goldsmith-Pinkham et al. 2020). In a long difference there is a single realisation of 33 origin shocks, and the shocks are heavy-tailed because some origins had few residents in 2003. Section 9 shows that shock-based inference (Adão et al. 2019; Borusyak and Hull 2023) is uninformative in this setting, so the identifying assumption is in practice that the 2003 shares of the origins that carry the instrument (Colombia, Peru and Romania, with 84 % of the Rotemberg weight) are unrelated to later changes in the entry gap, conditional on controls. I assess it with the sum-of-shares control recommended by Borusyak et al. (2022), by dropping each of these origins, with balance and pre-period tests, and with an annual panel in which shocks vary by year.

Prediction P3 is tested by interacting the inflow with standardised measures of geographic constraints and instrumenting the interaction with z_i times the constraint. Geographic constraints are predetermined and, unlike density or construction, do not reflect past demand.

8 Results

8.1 Where a demand shock goes

Table 2 and figure 4 report the effects of the demand shock. A cumulative inflow of 1 % of the population raises the quality-adjusted entry gap by 0.57 % (Anderson-Rubin set [0.11, 1.74]; first-stage F 11.0), the gap per m^2 by 0.50 % and the raw gap by 0.71 %. Combined with the response of the rent of all leases, 0.22 % [−0.09, 0.62], the implied response of entry rents is 0.83 % [0.25, 2.21]: entry rents respond several times more than the average rent of all leases, as prediction P1 implies.

The shock raises population by 0.82 % per point of inflow: most of the inflow adds to the population, and the rest is offset by fewer arrivals of other residents (section 8.2). It does not raise income per consumption unit (0.08, [−0.42, 0.97]) or income per person (−0.09). It raises household size by 1.26 % [0.69, 2.51] and income per household by 0.89 % [0.34, 2.14]. Prediction P2 holds: households absorb the shock partly by sharing dwellings, which raises household income without raising the income available per person. The affordability of entrants deteriorates by 0.75 % per point of inflow [−0.06, 1.74], while that of insiders does not change significantly (0.18). The dwelling stock does not respond (−0.26, [−0.81, 0.18]), and the falsification test passes: the inflow of 2015–2024 does not predict rent growth in 2012–2015 (−0.09, [−0.31, 0.28]).

8.2 Who adjusts

Prediction P4 concerns quantities. Table 3, panel A, uses moves between municipalities in 2021–2024, instrumented with the shift-share prediction for the same years. A foreign-born inflow of one point does

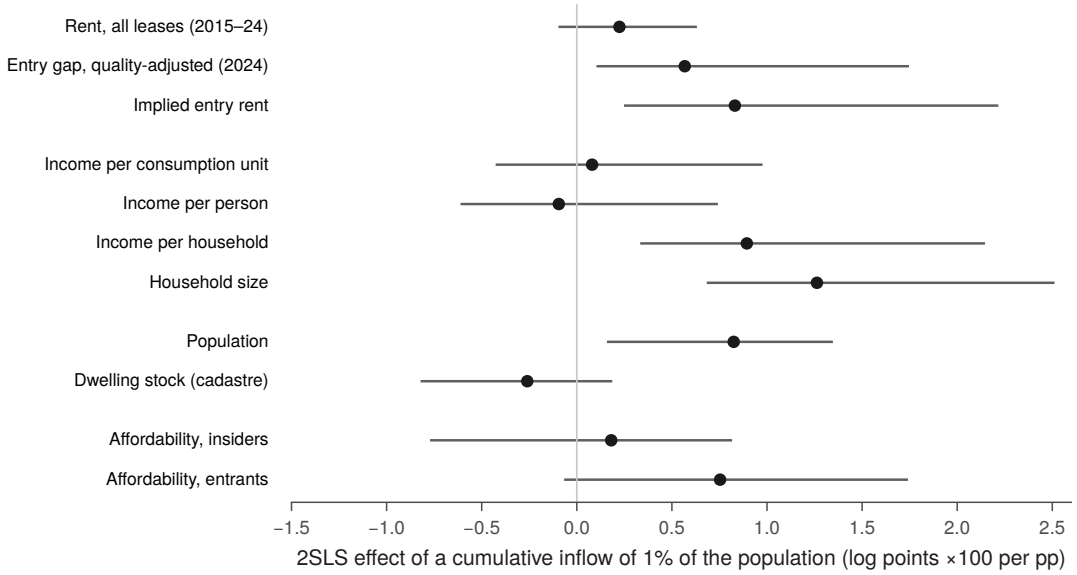


Fig. 4 Where a local demand shock goes

Notes: 2SLS coefficients of table 2 and Anderson-Rubin 95 % sets; arrows mark sets that extend beyond the axis.

Table 2 Where a local demand shock goes: rents, entry rents, incomes and households

Outcome	2SLS	SE	AR 95% set	First-stage F	N	Mean
A. Rents						
Rent, all leases, 2015–2024	0.22	(0.16)	[−0.09, 0.62]	21.0	695	0.200
Entry gap, per euro of cadastral value, 2024	0.57	(0.30)	[0.11, 1.74]	11.0	402	0.159
Entry gap, per m^2, 2024	0.50	(0.32)	[0.06, 1.90]	11.0	402	0.163
Entry gap, raw, 2024	0.71	(0.27)	[0.30, 1.73]	11.0	402	0.139
Implied entry rent (all leases + quality-adjusted gap)	0.83	(0.37)	[0.25, 2.21]	11.0	402	0.363
B. Incomes and households, 2015–2023						
Income per consumption unit (median)	0.08	(0.30)	[−0.42, 0.97]	16.0	695	0.310
Income per person	−0.09	(0.29)	[−0.60, 0.74]	16.0	695	0.318
Income per household	0.89	(0.37)	[0.34, 2.14]	16.0	695	0.293
Household size	1.26	(0.38)	[0.69, 2.51]	16.0	695	−0.022
C. Affordability						
Insiders: rent of all leases minus income per consumption unit	0.18	(0.34)	[−0.76, 0.81]	16.0	695	−0.144
Entrants: implied entry rent minus income per consumption unit	0.75	(0.37)	[−0.06, 1.74]	11.0	402	0.057
D. Quantities and falsification						
Population	0.82	(0.27)	[0.17, 1.34]	21.0	695	0.065
Dwelling stock (cadastre)	−0.26	(0.23)	[−0.81, 0.18]	21.0	695	0.038
Rent of all leases, 2012–2015 (pre-period)	−0.09	(0.13)	[−0.31, 0.28]	20.9	694	−0.035

Notes: one observation per municipality. Treatment: cumulative net foreign-born inflow between 1 January 2015 and 1 January 2025 (2024 for panels B and C, insiders) relative to 2015 population, instrumented with the corresponding sum of leave-own-province-out shift-share predictions (33 origins, 2003 settlement). All regressions include province fixed effects and predetermined controls (2011 census: log population, renting, vacancy, tertiary education, share aged 65+; 2012: construction, industry and trade-hospitality shares of firms, firms per capita; coastal and FUA-core indicators), population weights and standard errors clustered by province. Entry-gap outcomes are available for the municipalities above 20,000 inhabitants in the AEAT statistics. AR: Anderson-Rubin set.

not change the out-migration of Spanish nationals to other municipalities (-0.01, $[-0.40, 0.27]$, against a mean of 9.9 moves per 100 residents over the four years); it lowers their in-migration from other municipalities by -0.54 $[-2.00, -0.14]$, so that their net migration falls by -0.53 $[-1.82, -0.19]$. Foreign nationals move more in both directions. Incumbents stay; the households who would have moved in from elsewhere do not. Adjustment in quantities, like adjustment in prices, falls on entrants.

Table 3 Margins of adjustment: residential flows, heterogeneity, turnover and renters

A. Moves between municipalities per 100 residents, 2021–2024, per point of foreign-born inflow (2SLS)

	2SLS	SE	AR 95% set	F	Mean
Out-migration of Spanish nationals	−0.01	(0.13)	[−0.40, 0.27]	10.1	9.93
In-migration of Spanish nationals	−0.54	(0.31)	[−2.00, −0.14]	10.1	9.59
Net migration of Spanish nationals	−0.53	(0.27)	[−1.82, −0.19]	10.1	−0.34
Out-migration of foreign nationals	0.85	(0.22)	[0.55, 1.86]	10.1	4.09
In-migration of foreign nationals	0.60	(0.16)	[0.36, 1.28]	10.1	4.05

B. Heterogeneity: inflow and inflow × indicator (2SLS)

	Entry gap		Rent, all leases		Share
	Inflow	× indicator	Inflow	× indicator	
Coastal municipality	−0.05 (0.38)	0.59 (0.23)	0.21 (0.17)	0.02 (0.13)	46 %
More than 100,000 inhabitants	0.54 (0.45)	0.05 (0.48)	0.48 (0.27)	−0.45 (0.20)	57 %
Core city of a FUA	0.30 (0.51)	0.28 (0.42)	0.59 (0.29)	−0.42 (0.18)	75 %
Renting share in 2011 above mean	0.12 (0.50)	0.62 (0.54)	0.50 (0.23)	−0.29 (0.24)	47 %

C. Turnover, updates and renters

Annual exit rate of leases, Aragón register: all; Zaragoza	19.2 %; 15.7 %	median duration 3.3 years
Weight of new leases in the INE index: 2021; 2024	17.6 %; 12.3 %	
Entry gap (INE): 2021; 2024; 2024 without update caps	2.3; 12.7; 6.7	share of widening due to caps $\leq$ 58 %
Renters' median rent-to-income ratio (EPF): 2016; 2024	27.6 %; 26.4 %	share above 40 %: 25 %; 19 %
Rent per m^2 premium of foreign-national renters: 2016–18; change to 2022–24	−3.0 %; 3.5 (2.0)	young natives: change 0.6 (2.6)

Notes: panel A, INE Estadística de Migraciones y Cambios de Residencia (tables 69743 and 69745), moves between Spanish municipalities in 2021–2024 per 100 residents in 2021; treatment: foreign-born net inflow over the same years relative to 2021 population (points), instrumented with the shift-share prediction for 2021–2024; controls, fixed effects, weights and clusters as table 2. Panel B: 2SLS with the inflow and its interaction with the indicator, instrumented with the prediction and its interaction; same specification. Panel C: exit rate from $S(a) = p + (1-p)e^{-ha}$ fitted to the share of each deposit cohort (2012–2025) not yet returned; the update-cap accounting adds back to existing-lease growth the difference between CPI inflation and the cap for the share of each year in which the cap applied, an upper bound because not all leases are indexed to the CPI. EPF: INE household budget survey, households renting at market price (23,274 household-years, 2016–2024); premium from a weighted regression of log rent per m^2 on nationality and age of the reference person by period, log floor area and year, region and municipality-size fixed effects; robust standard errors in parentheses.

8.3 Supply constraints

Table 4 and figure 5 test prediction P3. A one-standard-deviation increase in the share of undevelopable land within 10 km raises the effect of the demand shock on the entry gap by 0.49 (p = 0.019) and on implied entry rents by 0.53 (p = 0.016). The interaction is similar for the sea (0.33, p = 0.021) and for steep slopes (0.78, p = 0.066), and it survives the inclusion of the interaction between the inflow and tourist exposure (0.48, p = 0.003). The geographic constraint does not change the response of the rent of all leases (0.03) or of the cadastral dwelling stock (0.02), which is close to zero everywhere. The amplification works through the entry margin, not through measured construction. Two caveats apply: the partial first-stage F of the interaction is modest (5.9 for the composite measure), and the marginal effect on the entry gap is only clearly positive above about 60 % of undevelopable land. Table 3, panel B, points the same way: the response of the entry gap is 0.59 larger (standard error 0.23) in coastal

municipalities, and it does not differ by size or between core cities and their peripheries. The response of the rent of all leases, by contrast, is smaller in large cities (−0.45, standard error 0.20) and in core cities (−0.42, 0.18), where leases are longer and turnover lower.

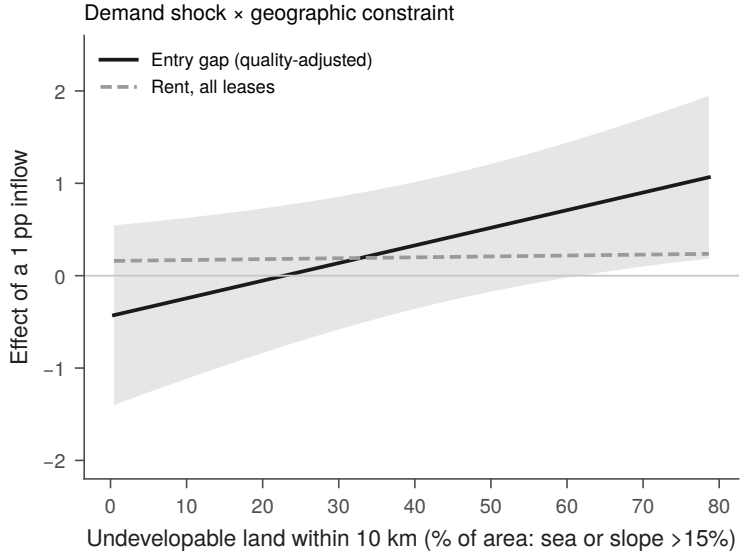


Fig. 5 Demand shocks and geographic constraints

Notes: marginal effect of a 1 pp inflow on the quality-adjusted entry gap and on the rent of all leases as a function of undevelopable land within 10 km (5th to 95th percentile), from table 4; 95 % band for the entry gap.

Table 4 Demand shocks and geographic constraints on supply

Outcome	Constraint	Inflow	Inflow × constraint	p	F (inflow; interaction)	Net of inflow × tourism	
						Inflow × constraint	Inflow × tourism
Entry gap (quality-adjusted)	Undevelopable	0.12 (0.37)	0.49** (0.21)	0.019	10.6; 5.9	0.48*** (0.17)	−0.45** (0.20)
	Sea	−0.02 (0.31)	0.33** (0.14)	0.021	26.4; 11.5	0.43*** (0.13)	−0.46** (0.19)
	Slope >15%	0.76 (0.50)	0.78* (0.42)	0.066	10.6; 9.5	0.60** (0.29)	−0.40 (0.29)
Implied entry rent	Undevelopable	0.34 (0.34)	0.53** (0.22)	0.016	10.6; 5.9	0.51*** (0.19)	−0.37** (0.17)
	Sea	0.19 (0.28)	0.36** (0.17)	0.029	26.4; 11.5	0.45*** (0.12)	−0.38** (0.15)
	Slope >15%	0.99 (0.53)	0.78* (0.41)	0.056	10.6; 9.5	0.58* (0.31)	−0.33 (0.27)
Entrants' affordability	Undevelopable	0.40 (0.30)	0.38* (0.21)	0.065	10.6; 5.9	0.33 (0.23)	0.08 (0.18)
	Sea	0.19 (0.29)	0.33* (0.19)	0.078	26.4; 11.5	0.30* (0.17)	0.08 (0.19)
	Slope >15%	0.84 (0.48)	0.52 (0.37)	0.165	10.6; 9.5	0.45 (0.39)	0.09 (0.24)
Rent, all leases	Undevelopable	0.19 (0.14)	0.03 (0.11)	0.814	16.1; 8.2		
	Sea	0.22 (0.11)	0.00 (0.05)	0.929	51.7; 16.5		
	Slope >15%	0.19 (0.18)	0.00 (0.16)	0.981	15.0; 12.5		
Dwelling stock	Undevelopable	−0.31 (0.25)	0.02 (0.09)	0.800	16.1; 8.2		
	Sea	−0.28 (0.29)	0.02 (0.07)	0.746	51.7; 16.5		
	Slope >15%	−0.32 (0.24)	0.00 (0.13)	0.997	15.0; 12.5		

Notes: 2SLS with two endogenous regressors, the inflow and its interaction with a standardised constraint, instrumented with the shift-share prediction and its interaction. Constraints: share of land within 10 km of the municipality centroid that is sea or has slope above 15 % (Copernicus DEM GLO-90), and each component. Same fixed effects, controls, weights and clustering as table 2. F: first-stage partial F statistics. The last two columns add the inflow interacted with the share of dwellings in tourist use (2020) and its instrument. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8.4 Tourism

The interaction between the inflow and tourist exposure in table 4 is negative (−0.45, p = 0.025): the same demand shock widens the entry gap less where many dwellings are tourist lets, as if part of that

stock acted as a reserve for long-term letting. The collapse of tourism in 2020–21 tests this margin directly (table 5, panel A; figure 6). In Catalan municipalities where 3–8 % and more than 8 % of dwellings were tourist lets, the number of new long-term leases rose by 5.5 % and 6.7 % during the collapse relative to municipalities with little tourist letting. With the recovery, rents on new leases rose by about 3 % in the same groups (3.1 % and 3.1 %). Below 3 % of dwellings there is no reallocation of leases.

A threshold regression (Hansen 2000) lets the data choose where the response changes. Allowing the slope of the collapse, recovery and later effects in tourist exposure to differ above and below a threshold, least squares locates it at 2.4 % of dwellings for new leases signed and for new-lease rents (likelihood-ratio set 2.4–3.2 % for rents); a wild cluster bootstrap rejects a common slope (p <0.002 for leases signed, 0.012 for rents). 41 of the 274 Catalan municipalities lie above it. Tourism shapes entry rents where tourist lets are a sizable share of the stock, as prediction P3 implies for a reallocation elasticity that is close to zero in markets with few tourist lets.

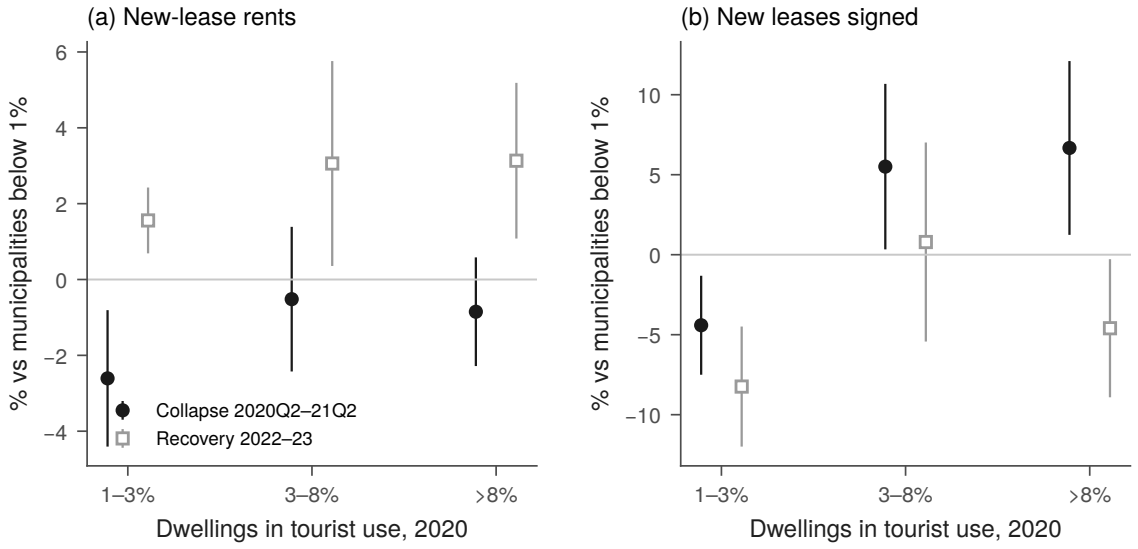


Fig. 6 Tourist dwellings and new leases during the collapse and recovery of tourism

Notes: coefficients and 95 % intervals of table 5, panel A, by share of dwellings in tourist use in August 2020.

8.5 Regulation

Prediction P5 says that caps on updates widen the entry gap and that caps on entry rents narrow it. Between 2021 and 2024 the national entry gap widened in every year, from 2.3 to 12.7 log points. In 2022 existing leases rose by 2.1 % while inflation was 8.1 %. Adding back to existing leases the difference between inflation and the cap, for the share of each year in which the cap applied (4.5 points in 2022 and 1.5 in 2023), would have left a gap of 6.7 points in 2024: the caps account for at most 58 % of the widening, an upper bound because not all leases are indexed to the CPI (table 3, panel C). The rest reflects faster growth of entry rents, which in 2024 rose by 8.8 % against 2.8 % for existing leases. Provincial

data do not show the decoupling the cap might have produced: across provinces, the growth of existing and of new leases was more correlated in 2022 (0.89) than in 2021 (0.76), so the cap held the level of existing-lease growth down without severing its link to local markets.

The Catalan cap on new leases, evaluated in the companion analysis with stacked difference-in-differences, lowered new-lease rents by about 5 % and the number of new leases by about 18 % in the first wave. Table 5, panel B, shows where part of those leases went: after the cap, the share of seasonal leases among new leases rose by 1.8 percentage points in the first wave (seasonal leases are exempt from the cap) and seasonal leases rose by about 26 %. In the whole sample their share rose from 6.6 % in 2023 to 11.1 % in 2025. The register of seasonal leases begins in 2023, so the pre-period is short (figure 7).

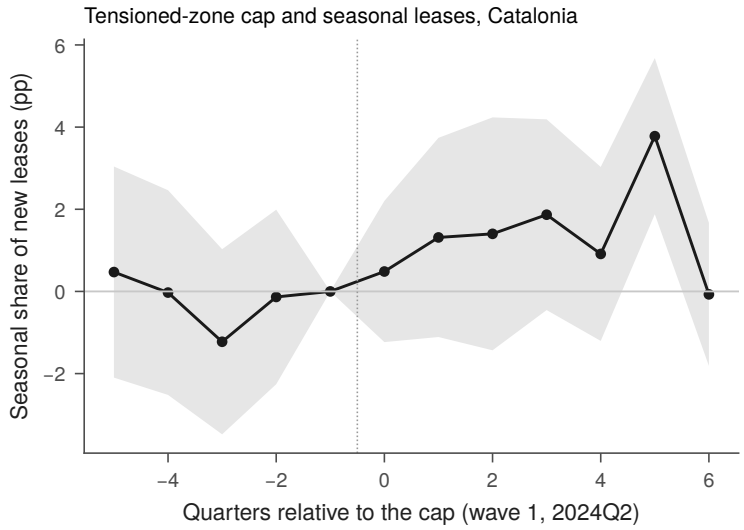


Fig. 7 Tensioned-zone cap and seasonal leases

Notes: event study of the seasonal share of new leases in the first wave of designated municipalities relative to never-designated ones, 2023Q1–2025Q4; 95 % band; clusters by municipality.

8.6 Tenure, spillovers and renters

If the rise in mortgage rates from 2022 had pushed would-be buyers into renting, rents should have risen more where housing was most expensive relative to incomes. Table 5, panel C, finds no break after 2022 in the index of all leases, and no association between price-to-income and the entry gap in 2024; the tenure channel is not supported by these data. Panel D studies spillovers within functional urban areas: the demand shock of the core city raises the dwelling stock of its peripheral municipalities (0.37, p = 0.040) but has small and imprecise effects on their rents and entry gaps. Moves tell the same story: the core's shock raises the in-migration of foreign nationals to the periphery (0.29 per point, p = 0.001) but not that of Spanish nationals (0.20, standard error 0.25). Supply in the periphery responds to demand in the core; rents there do not measurably follow.

The household budget survey does not identify recent movers, but households headed by a foreign national are likelier to have signed recently. Conditional on region, municipality size and floor area,

their rent per m^2 was $-3.0\,\%$ relative to other renters in 2016–2018 and rose by 3.5 points (standard error 2.0, $p = 0.08$) by 2022–2024, while that of young Spanish-headed households did not change (0.6, 2.6). The survey is consistent with an entry premium but, with these proxies, the evidence is weak.

Table 5 Tourism, regulation, tenure and spillovers

A. Tourist dwellings and new leases, Catalonia (% relative to municipalities with <1 % of dwellings in tourist use)

Exposure in 2020	New-lease rents			New leases signed		
	Collapse	Recovery	2024–25	Collapse	Recovery	2024–25
1–3 %	−2.6***	1.6***	−0.9	−4.4***	−8.2***	−12.4***
3–8 %	−0.5	3.1**	3.3***	5.5**	0.8	9.8
>8 %	−0.8	3.1***	2.3*	6.7**	−4.6**	8.8**

B. Tensioned-zone cap and seasonal leases, Catalonia, 2023Q1–2025Q4 (DiD)

	Seasonal share (pp)	log seasonal leases	log regular leases
Wave 1 (2024Q2)	1.82*** (0.39)	0.26*** (0.04)	−0.14*** (0.03)
Wave 2 (2024Q4)	0.94** (0.40)	0.09*** (0.03)	−0.08*** (0.03)

C. Tenure: price-to-income in 2021 (s.d.) × year, rent of all leases (reference 2021)

2019	2020	2022	2023	2024	Entry gap 2024
0.82***	0.21**	0.06	0.07	−0.01	−0.01

D. Spillovers: shift-share shock of the core city on peripheral municipalities of the same FUA (reduced form)

	Rent, all leases	Entry gap	Population	Inflow	Dwellings
Own shock	−0.04	0.13	−0.32**	−0.00	−0.20*
Core shock	0.13	0.29	0.39	0.03	0.37**

Notes: panel A, Catalan municipalities with new leases in every quarter of 2019; municipality and quarter fixed effects and exposure-bin × calendar-quarter terms; collapse 2020Q2–2021Q2, recovery 2022–2023; population weights; clusters by municipality. Panel B: municipality and quarter fixed effects and tourist exposure × calendar quarter; clusters by municipality; the seasonal-lease register starts in 2023. Panel C: IPVA municipalities with an appraised value (MIVAU, municipalities above 25,000 inhabitants); municipality and province × year fixed effects and predetermined characteristics × year; coefficients ×100. Panel D: 217 peripheral municipalities in 48 functional urban areas; province fixed effects and controls; clusters by FUA. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

9 Robustness

Shift-share identification. Table 6, panel A, adds the 2003 foreign-born share, scaled as the instrument, which Borusyak et al. (2022) recommend when shares do not sum to one: the entry gap responds by 0.64 [0.24, 3.64] and implied entry rents by 0.84 [0.30, 4.80], with a weaker first stage ($F = 5.6$); the rent of all leases responds by 0.18 [−0.04, 0.91]. Dropping from the instrument any of the five origins with the largest Rotemberg weights leaves the effect on the entry gap between 0.56 and 0.96, with Anderson-Rubin sets that exclude zero. The 2003 shares of Colombia and Peru are unrelated to rent growth in 2012–2015 conditional on the controls ($t = 0.5$ and -0.9), but the Romanian share is associated with slower pre-period growth ($t = -4.4$); without Romania the effect is unchanged.

Shock-based inference is not informative here. Recentring the instrument by permuting the 33 origin shocks across origins removes its first stage ($F = 0.08$), because the realised shocks are dominated by a few origins with small 2003 stocks; permuting within continents keeps a first stage ($F = 12.3$) but yields unbounded randomisation sets, and exposure-robust sets (Adão et al. 2019) are unbounded too.

The long-difference estimates therefore rest on the exogeneity of settlement shares (Goldsmith-Pinkham et al. 2020), with the usual caveat that past settlement may predict later inflows through persistent local conditions (Jaeger et al. 2018). A placebo by origin points the same way but has little power: inflows from high-income European origins, which more often buy than rent, do not raise the entry gap (-0.82, standard error 0.58) while the rest do (0.37, 0.24), but the first stage for the former is weak ($F = 1.3$).

Annual panel. Panel B uses the companion paper's annual panel, in which origin shocks vary by year and randomisation inference is informative, merged with annual rents at signature in Catalonia, Aragón and the Comunitat Valenciana (2,095 municipality-years, 221 municipalities). An inflow of 1 % raises new-lease rents by 0.53 and the rent of all leases by 0.00, so that the entry gap widens by 0.52. The estimates are imprecise ($F = 7.9$; randomisation p-values 0.079 for entry rents and 0.095 for the gap), and next year's predicted inflow is negatively related to this year's entry rents (-1.34, standard error 0.81), a sign that annual changes in rents at signature are noisy. The panel corroborates the direction of the cross-sectional results, not their magnitude.

Instrument, weights and registers. Panel C reports alternative instruments, weights and registers. With the companion paper's instrument, which sums annual predictions with centred timing over the 699 municipalities of its sample, the effect on the rent of all leases is larger (0.41) and the effect on the quality-adjusted entry gap is 0.43 [-0.03, 1.27]; the raw gap responds by 0.57 [0.15, 1.40]. Without population weights the first stage is weak (2.3 for the entry gap), and the estimates apply to the larger municipalities that hold most of the population. In the Basque register, which measures rents per m^2, the demand shock raises entry rents by 0.85 [-1.42, 2.26] and worsens entrants' affordability by 1.23 [0.30, 3.10]. The Catalan register illustrates why quality adjustment matters: its mean rent per contract falls with the demand shock (-0.55, [-1.25, -0.04]), consistent with arrivals shifting new leases towards cheaper and smaller dwellings, which the quality-adjusted tax measure removes.

Table 6 Robustness: identification, instrument, weights, registers and quality adjustment

	2SLS	SE	AR 95% set	F	N
A. Shift-share identification (long difference)					
Entry gap (quality-adjusted), controlling for the 2003 foreign-born share	0.64	(0.31)	[0.24, 3.64]	5.6	402
Implied entry rent, same control	0.84	(0.41)	[0.30, 4.80]	5.6	402
Rent of all leases, same control	0.18	(0.14)	[−0.04, 0.91]	7.4	695
Household size, same control	0.65	(0.21)	[0.35, 1.80]	7.7	695
Entry gap, instrument without Colombia	0.96	(0.55)	[0.24, 5.61]	5.8	402
Entry gap, instrument without Peru	0.68	(0.39)	[0.12, 2.30]	10.6	402
Entry gap, instrument without Romania	0.59	(0.38)	[0.03, 3.03]	6.4	402
B. Annual panel with year-varying shocks: Catalonia, Aragón and Comunitat Valenciana, 2012–2024					
New-lease rent, annual growth	0.53	(0.70)	$[-0.17, 1.41]^r$	7.9	2095
Rent of all leases, annual growth	0.00	(0.11)	$[-0.18, 0.24]^r$	7.9	2095
Change in the entry gap	0.52	(0.68)	$[-0.23, 1.35]^r$	7.9	2095
C. Instrument, weights and registers					
Companion-paper instrument (annual, centred timing), rent of all leases	0.41	(0.16)	[0.03, 0.75]	17.9	695
Companion-paper instrument, entry gap (quality-adjusted)	0.43	(0.29)	[−0.03, 1.27]	20.0	402
Companion-paper instrument, entry gap (raw)	0.57	(0.27)	[0.15, 1.40]	20.0	402
Unweighted, rent of all leases	0.25	(0.37)	[−0.36, 1.79]	8.7	695
Unweighted, entry gap (quality-adjusted)	1.09	(1.13)	$[-\infty, \infty]$	2.3	402
Basque Country, new-lease rent per m^2, 2016–2023	0.85	(0.75)	[−1.42, 2.26]	11.5	67
Basque Country, entrants' affordability (per m^2)	1.23	(0.59)	[0.30, 3.10]	17.2	63
Catalonia, mean new-lease rent (not quality-adjusted), 2015–2023	−0.55	(0.30)	[−1.25, −0.04]	61.4	288
Catalonia, income per consumption unit, 2015–2023	0.62	(0.44)	[−0.42, 1.38]	61.4	288

Notes: companion-paper instrument: sum of annual centred shift-share predictions over 2016–2024 for the 699 IPVA municipalities (Fernández-Aguilera 2026). Basque Country and Catalonia: long-difference instrument of this paper; controls are the 2011 log population, renting and vacancy shares; province fixed effects (Basque Country: three provinces, clusters by municipality). The Catalan deposit register reports the mean rent per contract, which is not adjusted for the size or quality of the dwelling. Panel A adds to the specification of table 2 the 2003 foreign-born share scaled as the instrument, the sum of exposure shares (Borusyak, Hull and Jaravel 2022), or drops from the instrument one of the three origins with the largest Rotemberg weights. Panel B: the panel of IPVA municipalities of the companion paper (annual inflow and instrument with centred timing, province × year fixed effects, 2003 foreign share and predetermined covariates × year, population weights) merged with annual new-lease rents (Catalonia: mean rent; Aragón: street-adjusted index; Comunitat Valenciana: median deposit; municipalities with at least 30 leases in the year); clusters by municipality. ^{r} Randomisation-inference set: origin shock paths permuted across origins (2,000 draws), recentred instrument, cluster-robust t statistic.

10 Discussion

What the evidence supports. Rental affordability in Spain deteriorated at entry and not for sitting tenants, in every source that observes rents at signature except the Basque register. Local demand shocks load on entry rents, more where geography constrains supply, while incomes per consumption unit do not respond and households grow larger. Incumbents do not leave; fewer households move in. Leases turn over slowly enough for the index of all leases to lag the market by years. Tourism and rent caps act on the same entry margin.

What it does not establish. The deterioration at entry is not universal: in the Basque Country, entry rents per m^2 grew less than incomes. The quality-adjusted entry gap is observed in levels only for 2024, so its response to demand rests on a cross-section and on the assumption that the gap was unrelated to the predicted inflow before the shock; the pre-period test on all leases supports, but does not prove, it. The long-difference instrument is identified from 2003 settlement shares dominated by three origins

rather than from as-good-as-random shocks, and the annual panel that uses shock variation is imprecise. Incomes are those of all households in the municipality, not those of entrants; the income of the households who sign new leases is not observed in open data. The tenure channel and spillovers on rents are not supported.

Welfare. A larger entry gap transfers resources from movers, young households and new arrivals to landlords and to sitting tenants, and it discourages mobility: households that would gain from moving for a job stay put, and those who need to move pay the reset. Larger households are an adjustment margin that the companion paper documents, not by itself a welfare loss. Caps on updates and longer leases protect insiders and widen the gap; caps on entry rents narrow it but reduce the number of leases offered on the regulated market and shift lets towards seasonal contracts.

11 Conclusion

The rise in Spanish rents is concentrated on households that enter the rental market. Official statistics show rents of sitting tenants growing more slowly than incomes everywhere; tax data and deposit registers show rents at signature growing faster than incomes in most markets and a quality-adjusted gap of about 16 log points between new leases and leases in force in 2024. The two facts are consistent: in a market of long, indexed leases, of which fewer than one in five turn over each year, the average rent of all leases absorbs only a fraction of what entrants pay, and with a lag of several years.

The paper explains the divergence with a stock-flow model in which local demand shocks are absorbed by rent resets, and tests its predictions with an immigration-driven demand shock. Entry rents respond several times more than the rents of all leases; incomes per consumption unit do not respond and households grow larger; incumbents stay and fewer households move in; the reset is larger where land is scarce and where tourism can withdraw dwellings from long-term letting; and regulation that protects incumbents widens the gap while caps at signature shift lets towards unregulated contracts. Prices and quantities adjust at entry.

The findings have three implications for policy. First, measurement: an index of all leases is the right deflator for what sitting tenants pay, but the wrong indicator of access. Statistics of rents at signature, adjusted for the quality and location of dwellings, exist in four regions and in the tax data for one year; publishing them annually for all of Spain would show where the problem is and how fast it grows. Second, supply: because the shock is absorbed at entry, measures that add dwellings to the flow available to entrants, through construction where land allows it and through the conversion of underused dwellings where it does not, act where the problem arises, while measures aimed at sitting tenants do not reach it. Third, regulation: each instrument moves the gap in a predictable direction. Caps on updates and longer minimum durations protect insiders and widen the gap; caps at signature narrow it in the regulated

segment but reduce the number of regulated leases and shift lets to segments the cap does not cover. A policy mix that protects tenants without closing the market to newcomers has to combine regulation with an expansion of the entry flow.

Two extensions would sharpen these conclusions. Linking successive deposits on the same dwelling would measure the reset directly and remove the need for quality adjustment, and observing the incomes of the households who sign new leases would measure the affordability of entrants rather than infer it from local averages. Both require administrative data that the registers already hold.

Declarations

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Competing interests The author declares no competing interests.

Data and code availability All data are public and are downloaded by the replication code from INE, the tax agency (AEAT), the Generalitat de Catalunya, the Gobierno de Aragón, the Gobierno Vasco, the Generalitat Valenciana, the Ministry of Housing, the European Space Agency (Copernicus DEM) and Eurostat.

Use of generative AI Claude (Anthropic) was used to retrieve the data and to assist with the code, the drafting and the English translation. The author verified all results and text and is fully responsible for the article.

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